4B 1 ALKANES FROM CRUDE OIL

SPECIFICATION REFERENCE 4.12

LEARNING OBJECTIVES

- Know that alkanes are used as fuels and obtained from the fractional distillation, cracking and reforming of crude oil, and be able to write equations for these reactions.

THE NEED FOR FUELS

The worldwide demand for energy is huge and steadily rising. At present, most of this energy comes from burning fossil fuels, in the form of coal, crude oil and natural gas. Most compounds in crude oil and natural gas are alkanes.



▲ **fig A** Petrol is just one product of crude oil, but is perhaps the best known.

In this topic we will look at the three main processes used to convert crude oil into fuels. They are:

- **fractional distillation**
- **cracking**
- **reforming.**

These processes are used in oil refineries located all over the world. You will already know something about fractional distillation, but you may be less familiar with cracking and reforming.



▲ **fig B** Several different processes take place in oil refineries. This distillation plant is just one small part of a large refinery.

FRACTIONAL DISTILLATION

Crude oil is a complex mixture of compounds, mostly hydrocarbons. The composition of the mixture varies quite a lot depending on which part of the world the crude oil comes from. The process is sometimes called 'fractionation' because it involves converting the crude oil into a small number of fractions. The number of fractions varies between different refineries but is typically six. Fractionation is done in a distillation column.

EXAM HINT

It is important to remember that a specific fraction from crude oil is still a mixture of compounds. It is just a smaller number of compounds within a defined range of boiling points.

THE PROCESS

The crude oil is first heated in a furnace, which turns most of it into vapour, which is then passed into the column near the bottom. There is a **temperature gradient** in the column: it is hotter near the bottom and cooler near the top.

As the vapour passes up the column through a series of bubble caps, different fractions condense at different heights in the column, depending on the boiling temperature range of the molecules in the fraction.

- Near the bottom of the column, the fractions contain larger molecules with longer chains and higher boiling temperatures.
- Near the top of the column, the fractions contain smaller molecules with shorter chains and lower boiling temperatures.
- Some of the hydrocarbons in crude oil are dissolved gases, and they rise to the top of the column without condensing.

Some fractions still contain many different compounds, so they may undergo further fractional distillation separately.

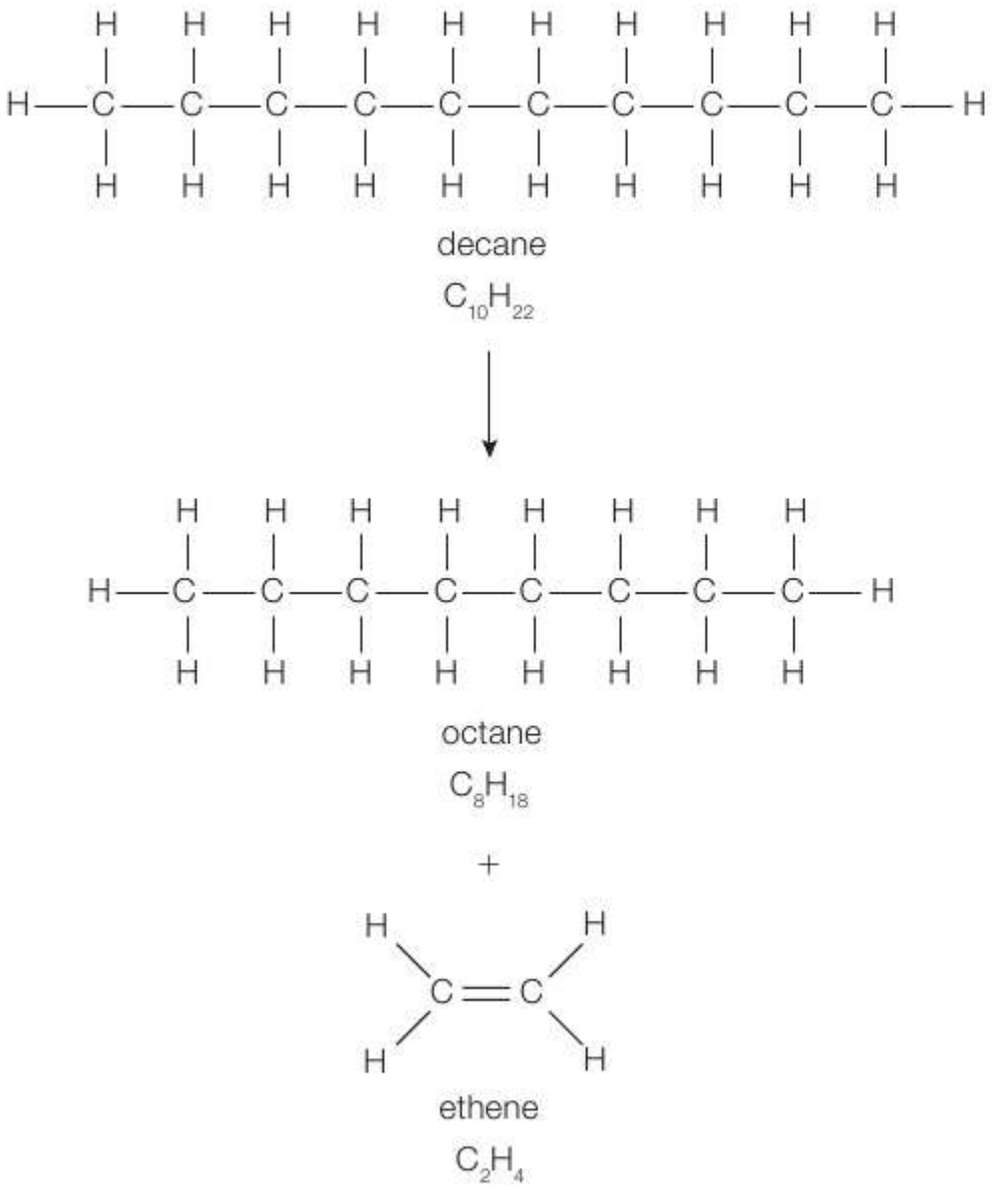
CRACKING

The world has fewer uses for longer-chain hydrocarbons so there is a surplus of these. The demand for shorter-chain hydrocarbons is much higher because they are better fuels and can be used to make other substances such as polymers. Unfortunately, there are not enough of these to satisfy the demand. The solution is to convert the longer chains into shorter chains, which is what happens in cracking.

THE PROCESS

Cracking is done by passing the hydrocarbons in the heavier (longer chain) fractions through a heated catalyst, usually of zeolite, which is a compound of aluminium, silicon and oxygen. This causes larger molecules to break up into smaller ones. From one large molecule, at least two smaller molecules are formed.

A good example is the cracking of decane into octane and ethene:



This is a good example because the two smaller molecules that are formed have familiar uses. Octane is one of the hydrocarbons in petrol and ethene is used to make polymers.

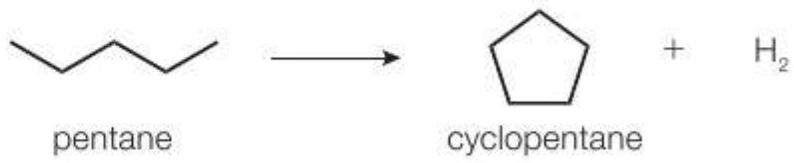
REFORMING

So far, we haven't mentioned one important point about the alkanes used as fuels. During the very rapid combustion that occurs in vehicle engines, not all hydrocarbons of the right size burn in the same way. Those with straight chains burn less efficiently than those with branched chains and those with rings (cyclic compounds). The process of reforming is used to convert straight-chain alkanes into branched-chain alkanes and cyclic hydrocarbons by heating them with a catalyst, usually platinum. This helps them to burn more smoothly in the engine.

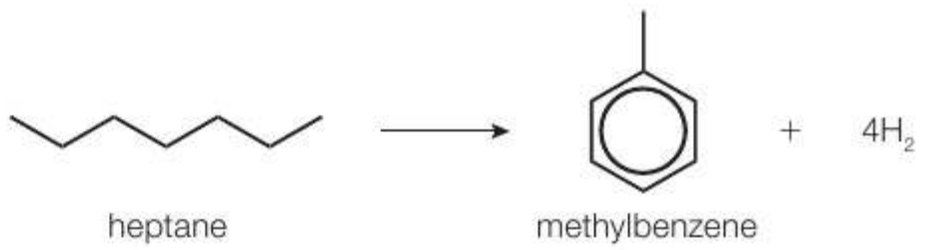
EXAMPLES

Here are some examples of reforming reactions, using skeletal formulae.

In the first one, pentane (C_5H_{12}) is converted into a cyclic alkane:



In the second one, heptane (C_7H_{16}) is converted into methylbenzene, which is a cyclic hydrocarbon but not an alkane. You will learn the meaning of the circle inside the hexagon later.



In each example, hydrogen is formed. It is a useful by-product.

LEARNING TIP

Be careful not to confuse fractional distillation and cracking.
Fractional distillation involves separating existing compounds, not making new ones in a chemical reaction.
Cracking involves a chemical reaction in which new compounds are formed.

CHECKPOINT

SKILLS: PROBLEM-SOLVING, ADAPTIVE LEARNING

1. One molecule of the alkane $C_{12}H_{26}$ is cracked to form two molecules of ethene and one molecule of a different alkane. What is the molecular formula of the alkane formed?
2. The products of a cracking reaction are two molecules of ethene and one molecule of pentane. What is the molecular formula of the alkane that is cracked?

SUBJECT VOCABULARY

fractional distillation the process used to separate a liquid mixture into fractions by boiling and condensing
cracking the breakdown of molecules into shorter ones by heating with a catalyst
reforming the conversion of straight-chain hydrocarbons into branched-chain and cyclic hydrocarbons
temperature gradient the way in which the temperature changes up and down the column

4B 2 ALKANES AS FUELS

SPECIFICATION REFERENCE
4.13 4.14 4.17(i)

LEARNING OBJECTIVES

- Know that pollutants, including carbon monoxide, oxides of nitrogen and sulfur, carbon particulates, and unburned hydrocarbons are emitted during the combustion of alkane fuels.
- Understand the problems arising from pollutants from the combustion of alkane fuels, limited to the toxicity of carbon monoxide and why it is toxic, and the acidity of oxides of nitrogen and sulfur.
- Understand the reactions of alkanes with oxygen in the air (combustion).

THE COMPLETE COMBUSTION OF ALKANES

As we mentioned earlier, alkanes can burn. They are burned in vast quantities to provide the world's energy. For example, propane is sold in containers at high pressure for use as a fuel, both in homes and when camping. The equation for its **complete combustion** is:

$$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$$

EXAM HINT

If you are asked to write an equation for the complete combustion of a hydrocarbon, balance the carbon and hydrogen atoms first and finish with the required amount of O_2. Do not include state symbols unless you are specifically asked to.

THE PRODUCTS OF COMBUSTION

One of the products of combustion is water, which is not a problem as it simply adds to the total quantity of global H_2O.

The other product is carbon dioxide. As you know, this is a greenhouse gas and most scientists consider its increasing production to be responsible for global warming, climate change and other problems.

EXAM HINT

Remember, carbon dioxide works by trapping infrared radiation that is emitted from the Earth's surface and so prevents it from escaping back into space.

Unfortunately, other problems are caused by using alkanes as fuels. The water and carbon dioxide formed are not considered to be pollutants by most people, but some other compounds formed during the combustion of alkanes are definitely pollutants.

INCOMPLETE COMBUSTION

Sometimes the combustion of an alkane is incomplete because there is not enough oxygen present, or because the combustion is very rapid. All of the hydrogen atoms in an alkane molecule are converted into water, but some of the carbon atoms can form gaseous carbon monoxide or solid carbon. These products can cause problems.

CARBON

You can often see when **incomplete combustion** forms solid carbon. This can be seen as smoke in the air or soot on the burner. One example of an equation for a reaction in which carbon is formed is:

$$C_3H_8 + 4O_2 \rightarrow C + 2CO_2 + 4H_2O$$

Notice that in this reaction two of the carbon atoms in propane undergo complete combustion and one does not. Tiny particles of carbon in the atmosphere can be harmful, but there is another product of combustion that can be fatal.

CARBON MONOXIDE

Carbon monoxide is a toxic gas that causes the death of many people each year. It acts by preventing the transport of oxygen around the body. It is colourless and odourless, so people breathe it into their lungs without knowing, which is why it is sometimes described as 'the silent killer'. Here is an example of an equation for a reaction in which carbon monoxide is formed:

$$C_3H_8 + 4O_2 \rightarrow 2CO + CO_2 + 4H_2O$$

UNBURNED HYDROCARBONS

The ultimate example of incomplete combustion is when the hydrocarbon does not burn at all. A small proportion of the hydrocarbons in a fuel are released into the atmosphere unchanged. They are known as unburned hydrocarbons (sometimes abbreviated to UHC).

OXIDES OF SULFUR

Some of the molecules in crude oil contain atoms of sulfur, and these may not be removed by the fractional distillation, cracking or reforming processes. During the combustion of alkanes, these atoms of sulfur form sulfur dioxide gas, and then can react in the atmosphere to form sulfur trioxide gas. The equations for these reactions are:

$$S + O_2 \rightarrow SO_2 \quad \text{and} \quad 2SO_2 + O_2 \rightarrow 2SO_3$$

Both sulfur gases are acidic oxides. When they dissolve in water in the atmosphere, they form sulfurous acid and sulfuric acid:

$$SO_2 + H_2O \rightarrow H_2SO_3 \quad \text{and} \quad SO_3 + H_2O \rightarrow H_2SO_4$$

Both acids contribute to the formation of acid rain. Acid rain is responsible for a lot of environmental damage, including damage to aquatic life in lakes and rivers, and damage to crops and forests.

OXIDES OF NITROGEN

Although very few molecules used as alkane fuels contain atoms of nitrogen, their combustion occurs at very high temperatures. Under these conditions, especially around the spark plugs in cars, these very high temperatures cause nitrogen molecules in the air to react with oxygen molecules. These reactions lead to the

formation of what are collectively known as oxides of nitrogen. They are represented by the formula NO_x. There are several of these oxides, but the main ones are nitrogen monoxide (NO) and nitrogen dioxide (NO_2).

At very high temperatures, the main reaction is:

$$N_2 + O_2 \rightarrow 2NO$$

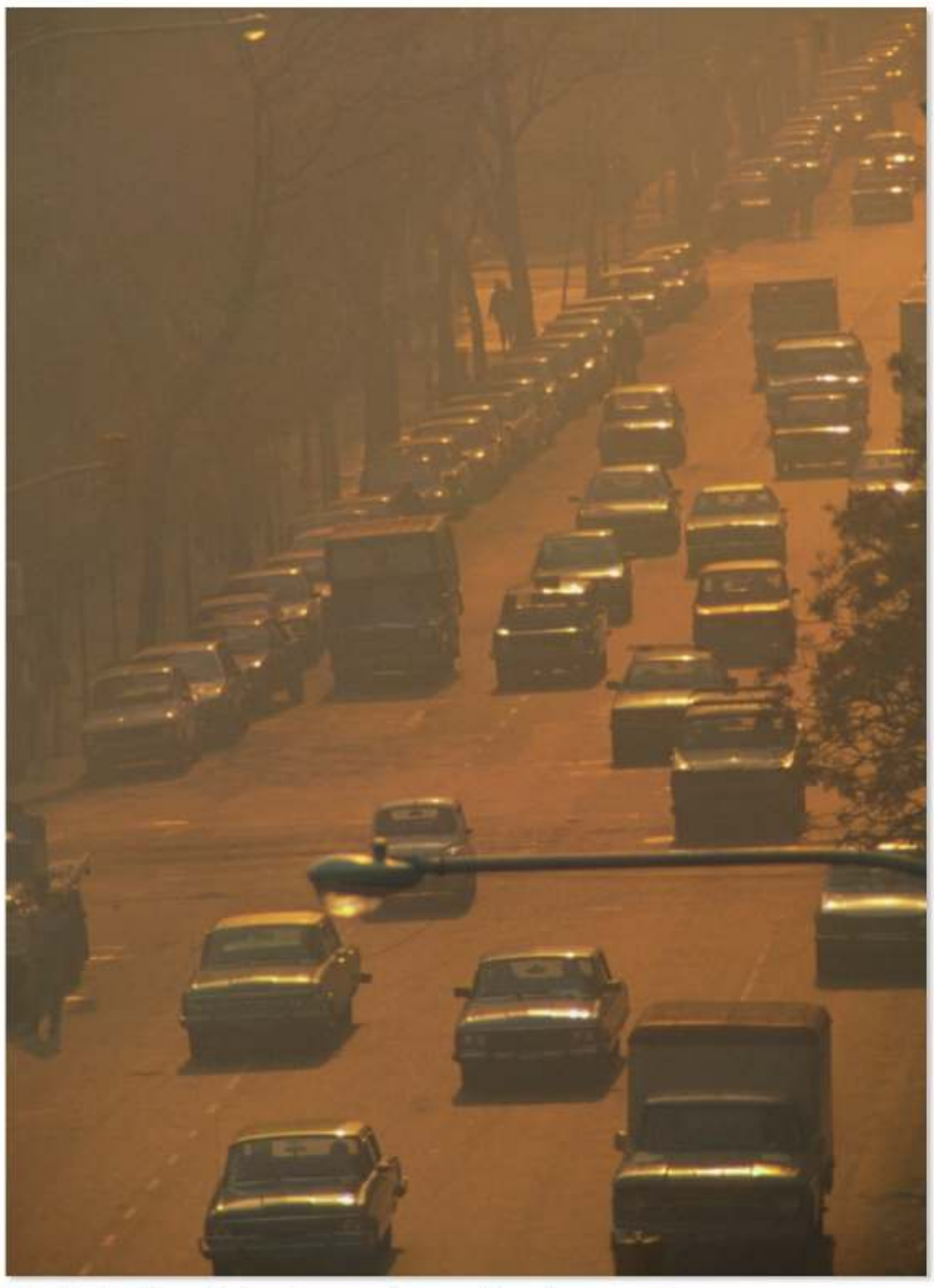
However, nitrogen monoxide can then react with more oxygen in the atmosphere as follows:

$$2NO + O_2 \rightarrow 2NO_2$$

Nitrogen dioxide is acidic and can dissolve in water in the atmosphere, forming nitrous acid and nitric acid:

$$2NO_2 + H_2O \rightarrow HNO_2 + HNO_3$$

Both acids contribute to environmental damage in the same way as sulfurous acid and sulfuric acid.

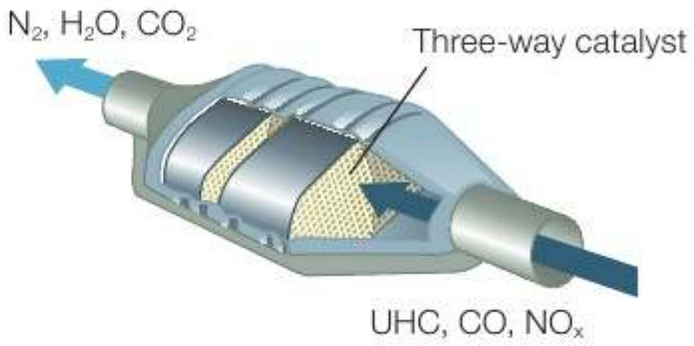


▲ **fig A** Air pollution is a growing problem in many cities.

CATALYTIC CONVERTERS TO THE RESCUE

Cars and other road vehicles are responsible for a lot of air pollution. The widespread use of catalytic converters fitted to exhaust systems has made pollution less of a problem.

There are different types of catalytic converter, but they all use small quantities of precious metals such as platinum, rhodium and palladium. These metals are spread thinly over a honeycomb mesh to increase the surface area for reaction (and also to save money). One common type is known as a three-way catalyst because it can remove three different pollutants: carbon monoxide, unburned hydrocarbons and oxides of nitrogen.



▲ **fig B** A three-way catalytic converter.

EXAM HINT

Also note that like many (but not all) catalysts, platinum, rhodium and palladium are found in the d-block of the Periodic Table.

As the exhaust gases from the engine pass through the catalytic converter, several reactions can occur.

Examples are the oxidation of the carbon monoxide and the oxidation of unburned hydrocarbons:

$$2CO + O_2 \rightarrow 2CO_2 \quad \text{and} \quad C_8H_{18} + 12\tfrac{1}{2}O_2 \rightarrow 8CO_2 + 9H_2O$$

Here is another useful reaction that removes two pollutants at the same time:

$$2NO + 2CO \rightarrow N_2 + 2CO_2$$

The catalysts currently used are not very good at removing sulfur compounds. The best way to prevent sulfur-based pollution is to remove the sulfur compounds from the fuel before the fuel is burned. This is done in some countries, where the resulting fuel is described as low sulfur or ultra-low sulfur fuel.

LEARNING TIP

Remember that even in the incomplete combustion of an alkane, all of the hydrogen atoms are completely oxidised to water.

CHECKPOINT

1. Summarise information about the products of combustion of alkanes, including names and whether complete or incomplete combustion was involved.

2. In a table, summarise the substances that react in a catalytic converter, and the products formed from them.

SUBJECT VOCABULARY

combustion a chemical reaction in which a compound reacts with oxygen

complete combustion reaction in which all of the atoms in the fuel are fully oxidised, producing only carbon dioxide and water

incomplete combustion reaction in which some of the atoms in the fuel are not fully oxidised, producing carbon dioxide, carbon monoxide and soot (unburnt carbon)

4B 3 ALTERNATIVE FUELS

SPECIFICATION REFERENCE
4.15 4.16

LEARNING OBJECTIVES

- Discuss the reasons for developing alternative fuels in terms of sustainability and reducing emissions, including the emission of CO_2 and its relationship to climate change.
- Apply the concept of carbon neutrality to different fuels, such as petrol, bioethanol and hydrogen.

THE NEED FOR ALTERNATIVE FUELS

There are serious concerns about relying on the combustion of fossil fuels to produce energy. We have already considered the pollution caused by the combustion of alkanes.

The other concerns are:

- the depletion of natural resources
- global warming and climate change.

Recently, there have been attempts to produce new fuels as alternatives to fossil fuels. Most of these fuels can be described as **biofuels**, which means that they are obtained from living matter that has died recently, rather than having died many millions of years ago. A wide definition of biofuels would include wood, which has been used as a fuel for many centuries and is still important in some countries today.

The terms 'renewable' and 'non-renewable' are often used when discussing energy sources. Non-renewable usually refers to coal, oil and natural gas. Renewable sources include biofuels, but also sunlight, wind, waves and tides, and geothermal energy.

CARBON NEUTRALITY

Fuels can be considered in terms of their carbon neutrality. Ideally, a fuel should be completely carbon neutral, although few are. The closer a fuel is to being carbon neutral, the better.

WHAT DOES CARBON NEUTRAL MEAN?

'**Carbon neutral**' is a term used to represent the idea of carbon dioxide neutrality. For example, when a tree grows, it absorbs carbon dioxide from the atmosphere, and the carbon atoms become part of the structure of the tree. If the tree is cut down and the wood is burned, then carbon dioxide is formed during its combustion. If the amount of carbon dioxide formed in the combustion is the same as the amount absorbed during the tree's growth, then the wood used is described as carbon neutral. This is because, over the time period between the tree starting to grow and the use of its wood as a fuel, the amount of carbon dioxide in the atmosphere has not been altered by its combustion.

You might imagine that fossil fuels, such as those formed from trees, could be described as carbon neutral because they too absorbed carbon dioxide during their growth and form the same amount of carbon dioxide when they are burned. The reason that they are not considered to be carbon neutral is that the carbon dioxide was absorbed from the atmosphere millions of years ago, when the amount of carbon dioxide in the atmosphere was much higher. When fossil fuels are burned, this increases the amount of carbon dioxide in today's atmosphere.

BIOALCOHOLS

You might think that **bioalcohols** are carbon neutral fuels because they are made from recently-grown plants that have absorbed carbon dioxide from the atmosphere. However, this does not recognise the fact that the plants have to be harvested, transported to a factory and processed in the factory, and the products transported to a point of sale. All these stages involve the use of energy, much of which involves the formation of carbon dioxide. Overall, the use of bioalcohols involves forming more carbon dioxide than is absorbed. Even so, bioalcohols are closer to being carbon neutral than fossil fuels.



▲ **fig A** Bioethanol (on the left) is sold next to unleaded petrol and diesel.

BIOETHANOL

Currently, the commonest bioalcohol is bioethanol. Remember that bioethanol and ethanol are not different compounds. Bioethanol is identical to ethanol. The 'bio' part of the name only refers to the method of production. For centuries, ethanol has been produced by the fermentation of sugars. This involves the use of yeasts that contain enzymes, but there is an upper limit to the concentration of the ethanol in the solution. The ethanol has to be separated from the much larger amount of water before it can be used as a fuel, and this separation requires energy.

It is now possible to use a wide range of plants, and also plant waste to produce ethanol. . The upper limit to the amount of ethanol that can be obtained from a given amount of starting material is increasing, and is much higher than in traditional fermentation. In the United States, corn is the main source of ethanol used in cars.



▲ **fig B** Not everyone agrees that it is a good idea for corn to be used as a fuel when it could be used to feed people.

EXAM HINT

Exam questions asking you to compare fuels usually carry 4 or more marks and are worth preparing on rough paper first before transferring to your exam paper. Avoid poorly presented and poorly structured answers.

COMPARING FUELS

BIOFUELS

The choice of alternative fuels is continually changing, as new sources of starting material and new processing methods are investigated. There are many factors to consider in any comparison, but for biofuels such as bioethanol these include the following.

- Land use – how much land is used to grow the crop? Should the land be used for other purposes, especially to grow food to feed people?
- Yield – how much of a crop can be grown on a given piece of land, and how quickly does it grow? What percentage of the carbon and hydrogen atoms in the crop ends up in the fuel?
- Manufacture and transport – how much energy is used in growing (including any fertilisers), processing and transporting the crop?
- Carbon neutrality – how close is the fuel to being carbon neutral?

HYDROGEN

For a long time, hydrogen has been considered an ideal alternative fuel, although it is not a biofuel. There are two main ways in which hydrogen can be used as a fuel in cars:

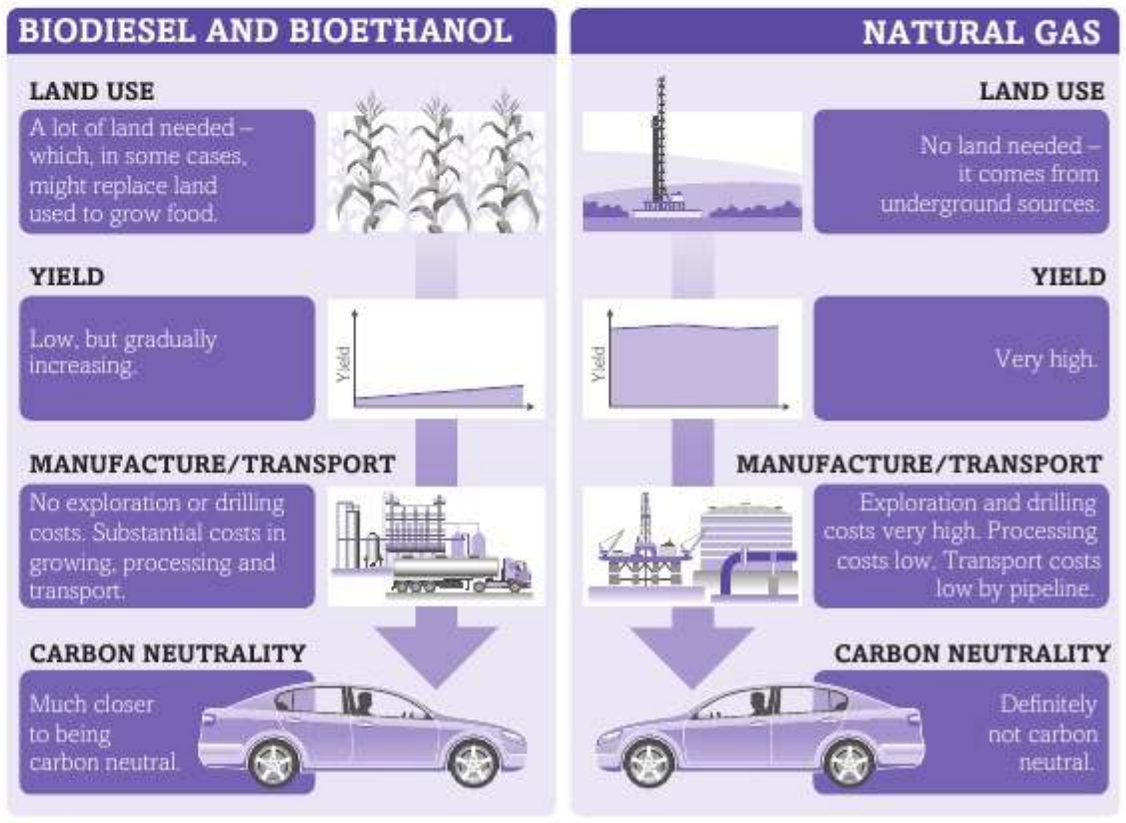
- it can be burned instead of a fossil fuel such as petrol or natural gas
- it can be used in a fuel cell to generate electricity that powers an electric motor.

Using a hydrogen fuel cell or burning hydrogen instead of a hydrocarbon in a car seems promising. No carbon dioxide is produced, which suggests no increase in the greenhouse effect. Hydrogen is very common, and is much more abundant than carbon in the Earth's crust. However, nearly all hydrogen is present in the water molecules in the oceans. Obtaining hydrogen gas from water is not difficult, but it requires energy, and where does the energy come from to do this? It could come from electricity, but how is the electricity generated? If the electricity comes from power stations that burn fossil fuels, then much of the advantage of using hydrogen instead of hydrocarbons is lost. Another major problem is hydrogen storage. The gas has a very low density, so storing enough inside a car needs a container under very high pressure. This means a very strong container and therefore one that is very heavy. It could be stored as a liquid, but as its boiling temperature is –253 °C, the container would need to be very cold and very well insulated.

Battery technology has improved greatly in recent years, and the range of battery-powered cars is continually increasing. Many scientists believe that this is a more practical alternative than using hydrogen as a fuel.

FOSSIL FUELS AND BIOFUELS

You should also be able to compare fossil fuels with biofuels. Here is one example of a comparison between such fuels.



▲ **fig C** Comparing bioethanol with natural gas as fuels for cars.

DID YOU KNOW?

One problem with using bioethanol as a fuel in vehicles is that internal combustion engines were designed to use petrol and not bioethanol. Currently very few cars use 100% bioethanol as their fuel. Normally the bioethanol sold is a blend of bioethanol and petrol, often identified by its E-number. Two common blends are E10 (10% bioethanol and 90% petrol) and E85 (85% bioethanol and 15% petrol).

LEARNING TIP

Focus on the bigger picture of comparing fuels, not on the chemical reactions that occur in their manufacture.

CHECKPOINT

SKILLS: REASONING, ARGUMENTATION

1. No carbon dioxide is formed when hydrogen is used as a fuel. Suggest why hydrogen is not a carbon neutral fuel.
2. Summarise reasons why a biofuel may not be carbon neutral.

SUBJECT VOCABULARY

biofuel fuel obtained from living matter that has died recently

bioalcohol fuel made from plant matter, often using enzymes or bacteria

carbon neutral a considered net zero effect on the amount of carbon dioxide in the atmosphere

4B 4 SUBSTITUTION REACTIONS OF ALKANES

SPECIFICATION REFERENCE
4.17(ii) 4.18

LEARNING OBJECTIVES

- Understand the reactions of alkanes with halogens.
- Understand the mechanism of the free radical substitution reaction between an alkane and a halogen.

WHAT IS A SUBSTITUTION REACTION?

You already know that the most common use of alkanes is as fuels. Combustion reactions are very important in producing energy, but are not very interesting from a chemist's point of view. This topic will give you a chance to increase your understanding of other reactions in organic chemistry.

Alkanes, apart from readily undergoing combustion, are fairly unreactive because they contain only carbon and hydrogen atoms and only single bonds. These bonds are also not very polar and so do not undergo reactions with substances that are considered to be very reactive, such as acids and alkalis and reactive metals.

There is a type of reaction that alkanes undergo, called a **substitution reaction,** which we will now look at in detail. Here is the equation for a reaction of the simplest alkane, methane:

$$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$$

You can see from the equation that one of the hydrogen atoms in methane has been replaced (substituted) by an atom of chlorine. The reaction can be described as chlorination or, in general (if another halogen were used), halogenation.

MECHANISMS

As you study organic chemistry more thoroughly, you will come across reactions that have been carefully studied, and for which there are explanations of exactly how they occur. The equation shown above for the reaction of methane with chlorine only shows the formulae of the reactants and products. It does not show how or why the reaction occurs.

A **mechanism** tries to explain the actual changes that occur during a reaction, especially in the bonding between the atoms. A mechanism is a sequence of two or more steps, each one represented by an equation, that shows how a reaction takes place.

THE CHLORINATION OF METHANE

When methane is only mixed with chlorine, no reaction occurs. If the temperature is increased, a reaction eventually occurs. However, the reaction will occur at room temperature if the mixture is exposed to ultraviolet radiation (or sunlight). We know that alkanes are not affected by ultraviolet radiation, but that ultraviolet radiation can affect chlorine. What happens in this reaction?

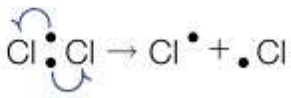
STEP 1

Ultraviolet radiation breaks the chlorine molecule into chlorine atoms. As the bond in the chlorine molecule consists of a shared pair of electrons, which are equally shared between the two atoms, then each chlorine atom takes one electron from the shared pair. This kind of bond breaking is called **homolytic fission**, a type of reaction you met in **Topic 4A.6**.

Step 1 can be represented in an equation:

$$Cl_2 \rightarrow Cl\bullet + Cl\bullet$$

However, it is better to use a different type of equation that shows clearly what happens to the electrons in the Cl–Cl bond.



The dots on the products each represent an unpaired electron. The formula Cl• does not represent an ion or a molecule; the term **free radical** (sometimes just radical) is used for it. A free radical is a species with an unpaired electron.

This equation above shows the original shared pair of electrons in the Cl–Cl bond. Each curly half-arrow shows what happens to the electrons. The upper arrow shows that one electron stays with the left-hand chlorine. The lower arrow shows that the other electron stays with the right-hand chlorine. Notice that the equation involves one molecule forming two free radicals. This type of reaction is called **initiation**, which means it starts the sequence of steps that forms the overall reaction.

STEP 2

Chlorine free radicals are very reactive species and when they collide with methane molecules they react by removing a hydrogen atom. An equation for this process is:

$$Cl\bullet + CH_4 \rightarrow HCl + CH_3\bullet$$

Notice that $CH_3\bullet$ is formed. This is a methyl free radical and, like Cl•, it is also very reactive. It can then react with chlorine molecules as follows:

$$CH_3\bullet + Cl_2 \rightarrow CH_3Cl + Cl\bullet$$

In this reaction, the methyl free radical removes a chlorine atom from a chlorine molecule, forming chloromethane and a chlorine free radical.

Notice that these two equations both involve one free radical reacting with one molecule, and that the products are also one free radical and one molecule. This type of reaction is called **propagation**, which means that the two steps considered together result in the conversion of CH_4 into the product CH_3Cl.

STEP 3

With all these free radicals being formed, it is likely that two of them will collide with each other. When this happens, they react to

form a molecule, as the two unpaired electrons are shared to form a covalent bond. As there are two different free radicals available, this means that there are three possibilities:

$Cl\bullet + Cl\bullet \rightarrow Cl_2$

$Cl\bullet + CH_3\bullet \rightarrow CH_3Cl$

$CH_3\bullet + CH_3\bullet \rightarrow C_2H_6$

These three equations all involve two free radicals reacting with each other to form one molecule. This type of reaction is called **termination**. This means that the sequence of reactions comes to an end because two reactive species are converted into unreactive species.

FURTHER SUBSTITUTION REACTIONS

You now know that a hydrogen atom in methane can be replaced by a chlorine atom in a substitution reaction. You can also see that the product chloromethane (CH_3Cl) still contains hydrogen atoms. These three hydrogen atoms can also be replaced, one by one, by chlorine atoms in similar substitution reactions.



▲ **fig A** Free radicals are important for understanding many chemical reactions, but free radicals in the human body can be harmful. Fortunately, antioxidants in fruit help protect the body from these harmful effects.

It is not easy to prevent these further substitution reactions from occurring. As well as the formation of chloromethane, these other reactions occur and other products are formed:

the formation of dichloromethane

$CH_3Cl + Cl_2 \rightarrow CH_2Cl_2 + HCl$

the formation of trichloromethane

$CH_2Cl_2 + Cl_2 \rightarrow CHCl_3 + HCl$

the formation of tetrachloromethane

$CHCl_3 + Cl_2 \rightarrow CCl_4 + HCl$

Each of these overall reactions can be represented by the same sequence of initiation, propagation and termination steps as for the formation of chloromethane.

The likelihood of these further reactions occurring means that this is not a good method for the preparation of chloromethane or other halogenoalkanes. This is because the yield will be low because of these further reactions, and also because several products have to be separated.

LEARNING TIP

Focusing on the three types of reaction will help you to understand radical mechanisms.

The three types are:

- initiation – one molecule becomes two free radicals
- propagation – a molecule and a free radical become a different free radical and molecule, and there are two reactions in this step
- termination – two free radicals become one molecule.

CHECKPOINT

1. Write the six equations for the mechanism of the reaction between methane and bromine.

2. Classify each of these reactions as initiation, propagation or termination. Explain your choice in each case.

(a) $C_2H_5\bullet + CH_3\bullet \rightarrow C_3H_8$

(b) $2O\bullet \rightarrow O_2$

(c) $F\bullet + CH_4 \rightarrow HF + CH_3\bullet$

SUBJECT VOCABULARY

substitution reaction reaction in which an atom or group is replaced by another atom or group

mechanism the sequence of steps in an overall reaction; each step shows what happens to the electrons involved in bond breaking or bond formation

homolytic fission the breaking of a covalent bond where each of the bonding electrons leaves with one species, forming a free radical

free radical a species that contains an unpaired electron

initiation first step that starts the reaction, involving the formation of free radicals, usually as a result of bond breaking caused by ultraviolet radiation

propagation the two steps that, when repeated many times, convert the starting materials into the products of a reaction

termination final step that involves the formation of a molecule from two free radicals, halting the reaction